

Single Variable Calculus

Personal Notes

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1 Derivatives, Slope, Velocity, and Rate of Change

Calculus links local change to global behavior. Its first local object is the derivative: geometrically, it is the slope of a tangent line; physically, it is an instantaneous rate such as velocity. Both interpretations come from the same limiting process.

1.1 Secant Slopes and the Derivative

Let $y = f(x)$ and fix a point $P = (x_0, f(x_0))$ on the graph. A nearby point $Q = (x_0 + \Delta x, f(x_0 + \Delta x))$ determines a *secant line*. Its slope is the *difference quotient*

$$\frac{\Delta f}{\Delta x} = \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}.$$

As Q approaches P , the secant line may approach a limiting line: the tangent line at P .

Definition 1.1 [*Derivative*]. If the limit exists, the derivative of f at x_0 is

$$f'(x_0) = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}.$$

It is the slope of the tangent line to the graph of f at x_0 .

The tangent line should not be defined merely as a line that meets the graph at one point: a line can cross a curve once without reflecting its local direction. The limiting-secants definition captures the direction of the curve near the point.

Note. The limit concerns nonzero values of Δx approaching zero. Substituting $\Delta x = 0$ into the difference quotient gives the meaningless expression $0/0$; algebraic simplification must come before taking the limit.

1.2 Example: the Reciprocal Function

Take $f(x) = 1/x$ at a point $x_0 \neq 0$. For $\Delta x \neq 0$,

$$\begin{aligned} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x} &= \frac{1}{\Delta x} \left(\frac{1}{x_0 + \Delta x} - \frac{1}{x_0} \right) \\ &= \frac{1}{\Delta x} \left(\frac{-\Delta x}{x_0(x_0 + \Delta x)} \right) \\ &= -\frac{1}{x_0(x_0 + \Delta x)}. \end{aligned}$$

The cancellation makes the limit visible:

$$f'(x_0) = -\frac{1}{x_0^2}.$$

The negative sign agrees with the graph: $1/x$ is decreasing on either side of the origin.

The tangent line through $(x_0, 1/x_0)$ is therefore

$$y - \frac{1}{x_0} = -\frac{1}{x_0^2}(x - x_0).$$

Setting $y = 0$ gives the x -intercept $2x_0$. By symmetry of $y = 1/x$ under exchanging x and y , the y -intercept is $2/x_0$. The triangle cut off by the coordinate axes therefore has the constant area $\frac{1}{2}|2x_0||2/x_0| = 2$.

1.3 Derivative Notation

Because several traditions developed alongside calculus, the derivative has several standard notations. Starting with

$$\Delta y = \Delta f = f(x_0 + \Delta x) - f(x_0),$$

the limiting relation can be written

$$\frac{\Delta y}{\Delta x} \longrightarrow \frac{dy}{dx} \quad \text{or} \quad \frac{\Delta f}{\Delta x} \longrightarrow f'(x_0).$$

Thus dy/dx , df/dx , f' , and Df all denote a derivative. The notation $f'(x_0)$ records the evaluation point explicitly; when using dy/dx , the point must be supplied by context.

1.4 The Power Rule from the Definition

For a positive integer n , let $f(x) = x^n$. The binomial expansion gives

$$(x + \Delta x)^n = x^n + nx^{n-1}\Delta x + (\Delta x)^2 R(x, \Delta x),$$

where R collects the remaining finitely many terms. Hence

$$\frac{(x + \Delta x)^n - x^n}{\Delta x} = nx^{n-1} + \Delta x R(x, \Delta x).$$

The second term vanishes as $\Delta x \rightarrow 0$, proving the *power rule* for monomials and, term by term, for polynomials:

$$\frac{d}{dx}x^n = nx^{n-1}.$$

1.5 Velocity as an Instantaneous Rate

The same difference quotient has a physical meaning. If $y(t)$ is the height of an object, then

$$\frac{\Delta y}{\Delta t}$$

is its average velocity over a time interval, while the limiting value $y'(t)$ is its *instantaneous velocity*.

For an object dropped from a height of 400 feet, the model

$$y(t) = 400 - 16t^2$$

places impact at $t = 5$ seconds because $y(5) = 0$. Its average downward speed over the full fall is $400/5 = 80$ feet per second. The instantaneous velocity is

$$y'(t) = -32t,$$

so at impact

$$y'(5) = -160 \text{ ft/s.}$$

The negative sign records direction: height is decreasing. Speed is the magnitude, 160 feet per second.

Note. The tangent-slope and velocity viewpoints are not separate definitions. Both are limits of average change over shorter and shorter intervals, which is why one derivative describes geometry and motion at once.

Source: [MIT 18.01 Single Variable Calculus](#), Fall 2006, Lecture 1, "Derivatives, Slope, Velocity, and Rate of Change," taught by David Jerison.